

Lesson 01 · Quantities, percentages and algebra

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Without using a calculator, evaluate $10 \times (-2)$.

☐ A -20

☐ B 20

☐ C -12

☐ D 12

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Q2 · Rank 1 / Foundation

Without using a calculator, evaluate $7 - (-4)$.

☐ A 3

☐ B -11

☐ C 11

☐ D -3

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Lesson 01 · Quantities, percentages and algebra

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
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Q3 · Rank 1 / Foundation

Without using a calculator, find the value of $2x - y$ when $x = 7$ and $y = 4$.

☐ A 6

☐ B 18

☐ C 3

☐ D 10

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Q4 · Rank 2 / Core

Simplify the following algebraic expression by collecting like terms: $6x + 5y - 2x - 12y$.

☐ A $4x - 7y$

☐ B $4x + 17y$

☐ C $8x - 7y$

☐ D $-3xy$

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Lesson 01 · Quantities, percentages and algebra

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Q5 · Rank 2 / Core

Reduce the numerical fraction $\frac{18}{30}$ to its lowest terms.

☐ A $\frac{9}{10}$

☐ B $\frac{3}{5}$

☐ C $\frac{5}{3}$

☐ D $\frac{2}{5}$

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Q6 · Rank 2 / Core

Calculate 5% of 24.

☐ A 12

☐ B 0.12

☐ C 1.2

☐ D 120

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Lesson 01 · Quantities, percentages and algebra

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Q7 · Rank 3 / Application

In 2011 in the United States, 35 out of every 100 adults owned a smartphone. By 2013 this figure increased to 56 out of every 100. By what factor did smartphone ownership increase during this period? Give your answer as a mixed fraction in its lowest terms.

☐ A $\frac{5}{8}$ ☐ B $1\frac{3}{5}$ ☐ C $\frac{3}{5}$ ☐ D $2\frac{1}{10}$

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Q8 · Rank 3 / Application

A firm has 132 female and 88 male employees. What percentage of staff are female?

☐ A 40%☐ B 66%☐ C 150%☐ D 60%

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Q9 · Rank 3 / Application

A student discount card reduces a bill in a restaurant from USD 124 to USD 80.60. Work out the percentage discount.

☐ A 35%☐ B 65%☐ C 53.85%☐ D 43.40%

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Q10 · Rank 4 / Challenge · OPTIONAL

A firm has 132 female and 88 male employees. During the next year 8 additional female staff are employed. If the percentage of female staff is now 56%, how many additional male staff were recruited during the year?

☐ A 22☐ B 30☐ C 8☐ D 12

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Lesson 02 · Functions, lines and market equilibrium

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

If $f(x) = 3x + 15$, evaluate $f(2)$.

☐ A 6

☐ B 17

☐ C 21

☐ D 36

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Q2 · Rank 1 / Foundation

If $f(x) = 3x + 15$, evaluate $f(0)$.

☐ A 0

☐ B 3

☐ C 18

☐ D 15

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Lesson 02 · Functions, lines and market equilibrium

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

For the supply function $P = \frac{1}{3}Q + 7$, determine P when $Q = 12$.

☐ A 11☐ B 4☐ C 43☐ D 19

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Q4 · Rank 2 / Core

State the value of the slope and y intercept for the line $y = 5x + 9$.

☐ A Slope 9; intercept 5☐ B Slope 5; intercept 9☐ C Slope -5; intercept 9☐ D Slope 5; intercept -9

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Lesson 02 · Functions, lines and market equilibrium

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

For the line $3x - 5y = 8$, find the value of x when $y = 2$.

☐ A $-\frac{2}{3}$

☐ B $\frac{8}{3}$

☐ C 6

☐ D 18

Jacques, 9th ed. · §1.3 · Exercise 1.3 4(a) · printed p. 52 / PDF p. 69

Q6 · Rank 2 / Core

A taxi firm charges a fixed cost of USD 4 plus a charge of USD 2.50 a mile. Write down a formula for the cost, C , of a journey of x miles.

☐ A $C = 6.5x$

☐ B $C = 4x + 2.5$

☐ C $C = 2.5x$

☐ D $C = 4 + 2.5x$

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Lesson 02 · Functions, lines and market equilibrium

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

An airline charges USD 300 for a flight of 2000 km and USD 700 for a flight of 4000 km. Assuming a linear model, estimate the cost of a flight of 3200 km.

☐ A 480☐ B 640☐ C 540☐ D 500

Jacques, 9th ed. · §1.3 · Exercise 1.3 2(b)(i) · printed p. 52 / PDF p. 69

Q8 · Rank 3 / Application

A taxi firm charges a fixed cost of USD 4 plus a charge of USD 2.50 a mile. Hence, or otherwise, work out the distance of a journey which costs USD 24.

☐ A 8 miles☐ B 9.6 miles☐ C 10 miles☐ D 11.2 miles

Jacques, 9th ed. · §1.3 · Exercise 1.3 9(c) · printed p. 52 / PDF p. 69

Lesson 02 · Functions, lines and market equilibrium

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

If the supply function is $Q = 4P - 25$, determine the lowest possible price that could be charged.

☐ A 25☐ B 0☐ C $\frac{25}{4}$ ☐ D $\frac{4}{25}$

Jacques, 9th ed. · §1.5 · Chapter 1 multiple choice Question 3 · printed p. 112 / PDF p. 129

Q10 · Rank 4 / Challenge · OPTIONAL

The demand and supply functions of a good are given by $P = -3Q_D + 75$ and $P = 2Q_S + 15$. Determine the equilibrium price.

☐ A 39☐ B 12☐ C 60☐ D 15

Jacques, 9th ed. · §1.4–1.5 · Chapter 1 multiple choice Question 6 · printed p. 113 / PDF p. 130

Lesson 03 · Quadratics, revenue, cost and profit

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Solve the quadratic equation $x^2 = 81$.

☐ A $x = 9$ only

☐ B $x = -9$ only

☐ C $x = \pm 81$

☐ D $x = \pm 9$

Jacques, 9th ed. · §2.1 · Exercise 2.1 1(a) · printed p. 137 / PDF p. 154

Q2 · Rank 1 / Foundation

Solve $f(x) = 0$ for $f(x) = x(100 - x)$.

☐ A $x = 0$ or $x = 100$

☐ B $x = 100$ only

☐ C $x = 0$ only

☐ D $x = -100$ or $x = 100$

Jacques, 9th ed. · §2.1 · Exercise 2.1 4(b) · printed p. 137 / PDF p. 154

Lesson 03 · Quadratics, revenue, cost and profit

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

If the demand function of a good is given by $P = 80 - 3Q$, find the price when $Q = 10$, and deduce the total revenue.

☐ A $P = 50, TR = 50$

☐ B $P = 50, TR = 500$

☐ C $P = 77, TR = 770$

☐ D $P = 110, TR = 1100$

Jacques, 9th ed. · §2.2 · Exercise 2.2 1(a) · printed p. 148 / PDF p. 165

Q4 · Rank 2 / Core

Given the demand function $P = 10 - 4Q$, express TR as a function of Q .

☐ A $TR = 10 - 4Q$

☐ B $TR = 10Q - 4Q$

☐ C $TR = 10Q - 4Q^2$

☐ D $TR = 10Q + 4Q^2$

Jacques, 9th ed. · §2.2 · Exercise 2.2 2(c) · printed p. 148 / PDF p. 165

Lesson 03 · Quadratics, revenue, cost and profit

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

If fixed costs are 100 and variable costs are 5 per unit, find the total cost when $Q = 10$.

☐ A 50☐ B 105☐ C 1000☐ D 150

Jacques, 9th ed. · §2.2 · Exercise 2.2 1(b) · printed p. 148 / PDF p. 165

Q6 · Rank 2 / Core

For demand $P = 80 - 3Q$, fixed costs 100 and variable costs 5 per unit, use the revenue and cost when $Q = 10$ to work out the corresponding profit.

☐ A 350☐ B 500☐ C 150☐ D 650

Jacques, 9th ed. · §2.2 · Exercise 2.2 1(c) · printed p. 148 / PDF p. 165

Lesson 03 · Quadratics, revenue, cost and profit

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

Given the total revenue function $TR = 50Q - 4Q^2$, find the corresponding demand function. Assume $Q > 0$.

☐ A $P = 50Q - 4Q^2$

☐ B $P = 50 - 4Q$

☐ C $P = 50 - 8Q$

☐ D $P = 50 + 4Q$

Jacques, 9th ed. · §2.2 · Exercise 2.2 3(a) · printed p. 148 / PDF p. 165

Q8 · Rank 3 / Application

The total cost, TC , of producing 100 units of a good is 600 and the total cost of producing 150 units is 850. Assuming that the total cost function is linear, find an expression for TC in terms of Q , the number of units produced.

☐ A $TC = 6Q$

☐ B $TC = 5Q$

☐ C $TC = 100 + 5Q$

☐ D $TC = 500 + Q$

Jacques, 9th ed. · §2.2 · Exercise 2.2 6 · printed p. 148 / PDF p. 165

Lesson 03 · Quadratics, revenue, cost and profit

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

Given that fixed costs are 500 and variable costs are 10 per unit, express AC as a function of Q . Assume $Q > 0$.

☐ A $AC = 500 + 10Q$

☐ B $AC = 500/Q$

☐ C $AC = 10Q$

☐ D $AC = 500/Q + 10$

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Q10 · Rank 4 / Challenge · OPTIONAL

Given the quadratic supply and demand functions $P = Q_S^2 + 2Q_S + 12$ and $P = -Q_D^2 - 4Q_D + 68$, determine the equilibrium price and quantity. Use nonnegative quantity.

☐ A $P = 36, Q = 4$

☐ B $P = 27, Q = -7$

☐ C $P = 4, Q = 36$

☐ D $P = 68, Q = 0$

Jacques, 9th ed. · §2.1 · Exercise 2.1 9 · printed p. 137 / PDF p. 154

Lesson 04 · Powers, logarithms and exponential growth

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Without using your calculator, evaluate 8^2 .

☐ A 16

☐ B 64

☐ C 10

☐ D 256

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Q2 · Rank 1 / Foundation

Without using your calculator, evaluate 3^{-1} .

☐ A -3

☐ B 3

☐ C $\frac{1}{3}$

☐ D $-\frac{1}{3}$

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Lesson 04 · Powers, logarithms and exponential growth

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

Without using your calculator, evaluate $36^{1/2}$.

☐ A 18

☐ B 72

☐ C 1296

☐ D 6

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Q4 · Rank 2 / Core

Use the rules of indices to simplify $a^3 \times a^8$.

☐ A a^{11}

☐ B a^{24}

☐ C $2a^{11}$

☐ D a^5

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Lesson 04 · Powers, logarithms and exponential growth

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

Use the rules of logs to expand $\ln(xy)$. Assume $x > 0$ and $y > 0$.

☐ A $\ln x \ln y$

☐ B $\ln x + \ln y$

☐ C $\ln x - \ln y$

☐ D $x + y$

Jacques, 9th ed. · §2.4 · Exercise 2.4 2(a) · printed p. 183 / PDF p. 200

Q6 · Rank 2 / Core

Solve $e^x = 5.9$. Round your answer to two decimal places.

☐ A 5.90

☐ B 0.59

☐ C 1.77

☐ D -1.77

Jacques, 9th ed. · §2.4 · Exercise 2.4 4(a) · printed p. 183 / PDF p. 200

Lesson 04 · Powers, logarithms and exponential growth

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

The number of items, N , produced each day by an assembly-line worker, t days after an initial training period, is modelled by $N = 100 - 100e^{-0.4t}$. Calculate the number of items produced daily 1 day after the training period. Choose the value to two decimal places.

☐ A 67.03☐ B 100☐ C 40.00☐ D 32.97

Jacques, 9th ed. · §2.4 · Exercise 2.4 1(1)(a) · printed p. 182 / PDF p. 199

Q8 · Rank 3 / Application

For the worker model $N = 100 - 100e^{-0.4t}$, what is the worker's daily production in the long run?

☐ A 100 items/day☐ B 0 items/day☐ C 40 items/day☐ D Unlimited growth

Jacques, 9th ed. · §2.4 · Exercise 2.4 1(2) · printed p. 183 / PDF p. 200

Lesson 04 · Powers, logarithms and exponential growth

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

Solve $\ln x = 0$. Which x intercept does this give for $y = \ln x$?

☐ **A** $x = 0$

☐ **B** $x = 1$

☐ **C** $x = e$

☐ **D** $x = -1$

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Q10 · Rank 4 / Challenge · OPTIONAL

Future sales of two products A and B are given by $S_A = 5e^{0.01t}$ and $S_B = 2e^{0.02t}$. Find the time, t , when sales of the two products are the same.

☐ **A** $\ln(2.5)$

☐ **B** $100 \ln(0.4)$

☐ **C** $100 \ln(2.5)$

☐ **D** $50 \ln(2.5)$

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Lesson 05 · From average slope to the derivative

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Find the slope of the straight line passing through $(2, 5)$ and $(4, 9)$.

☐ A $\frac{1}{2}$

☐ B 2

☐ C 4

☐ D -2

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Q2 · Rank 1 / Foundation

Find the slope of the straight line passing through $(3, -1)$ and $(7, -5)$.

☐ A 1

☐ B -4

☐ C -1

☐ D 4

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Lesson 05 · From average slope to the derivative

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

Find the slope of the straight line passing through $(7, 19)$ and $(4, 19)$.

☐ A Undefined

☐ B -3

☐ C 19

☐ D 0

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Q4 · Rank 2 / Core

Differentiate the function $f(x) = x^7$.

☐ A $7x^6$

☐ B x^6

☐ C $7x^7$

☐ D $x^8/8$

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Lesson 05 · From average slope to the derivative

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

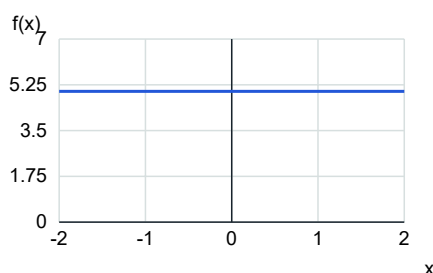
Differentiate $y = x^8$.

☐ A $8x^8$
☐ B $8x^7$
☐ C x^7
☐ D $x^9/9$

Jacques, 9th ed. · §4.1 · Exercise 4.1 5(a) · printed p. 277 / PDF p. 294

Q6 · Rank 2 / Core

Sketch the graph of the function $f(x) = 5$. Explain why it follows from this that $f'(x) = 0$. Choose the explanation.


☐ A The graph is vertical.

☐ B Its height is zero.

☐ C The graph is horizontal, so every tangent has zero slope.

☐ D Its slope is 5.

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Lesson 05 · From average slope to the derivative

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

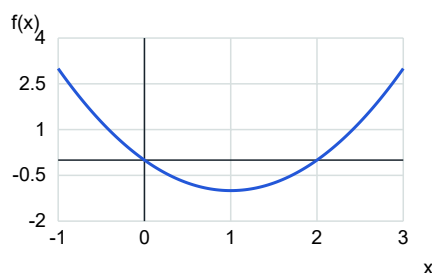
Differentiate the function $f(x) = x^7$. Hence calculate the slope of the graph of $y = x^7$ at the point $x = 2$.

☐ A 128☐ B 896☐ C 14☐ D 448

Jacques, 9th ed. · §4.1 · Exercise 4.1 4, hence instruction · printed p. 277 / PDF p. 294

Q8 · Rank 3 / Application

For the function $f(x) = x^2 - 2x$, sketch the graph and, by measuring the slope of the tangents, estimate $f'(1)$.

☐ A 0☐ B -1☐ C 2☐ D 1

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Lesson 05 · From average slope to the derivative

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

For the graph $y = \sqrt{x}$, A is the point where $x = 4$, and B is the point where $x = 4.1$. Find the gradient of the chord AB.

☐ A $\frac{1}{4}$

☐ B $\frac{\sqrt{4.1}-2}{4.1}$

☐ C $\frac{\sqrt{4.1}-2}{0.1}$

☐ D $\sqrt{0.1}$

Jacques, 9th ed. · §4.1 · Exercise 4.1* 3(a)(ii) · printed p. 278 / PDF p. 295

Q10 · Rank 4 / Challenge · OPTIONAL

Find the coordinates of the point(s) at which the curve $y = x^5$ has gradient 405.

☐ A (3, 243) only

☐ B (3, 243) and $(-3, -243)$

☐ C $(81, 81^5)$ only

☐ D (3, 405) and $(-3, 405)$

Jacques, 9th ed. · §4.1 · Exercise 4.1* 4(b) · printed p. 278 / PDF p. 295

Lesson 06 · Power, constant, sum and difference rules

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Differentiate $y = 5x^2$.

☐ A $5x$

☐ B $10x^2$

☐ C $10x$

☐ D x^2

Jacques, 9th ed. · §4.2 · Exercise 4.2 1(a) · printed p. 286 / PDF p. 303

Q2 · Rank 1 / Foundation

Differentiate $y = 2x + 3$.

☐ A $2x$

☐ B 5

☐ C 3

☐ D 2

Jacques, 9th ed. · §4.2 · Exercise 4.2 1(c) · printed p. 286 / PDF p. 303

Lesson 06 · Power, constant, sum and difference rules

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

Differentiate $y = ax + b$, treating a and b as constants.

☐ A a

☐ B b

☐ C $a + b$

☐ D ax

Jacques, 9th ed. · §4.2 · Exercise 4.2 1(h) · printed p. 286 / PDF p. 303

Q4 · Rank 2 / Core

Differentiate $y = x^2 + x + 1$.

☐ A $2x + 2$

☐ B $2x + 1$

☐ C $x + 1$

☐ D $2x$

Jacques, 9th ed. · §4.2 · Exercise 4.2 1(d) · printed p. 286 / PDF p. 303

Lesson 06 · Power, constant, sum and difference rules

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

Differentiate $y = x^2 - 3x + 2$.

☐ A $2x + 3$

☐ B $2x - 1$

☐ C $2x - 3$

☐ D $x^2 - 3$

Jacques, 9th ed. · §4.2 · Exercise 4.2 1(e) · printed p. 286 / PDF p. 303

Q6 · Rank 2 / Core

Differentiate $y = 2x^3 - 6x^2 + 49x - 54$.

☐ A $6x^2 - 6x + 49$

☐ B $6x^2 - 12x - 5$

☐ C $2x^2 - 6x + 49$

☐ D $6x^2 - 12x + 49$

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Lesson 06 · Power, constant, sum and difference rules

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

Evaluate $f'(x)$ for $f(x) = 3x^9$ at $x = 1$.

☐ A 27

☐ B 3

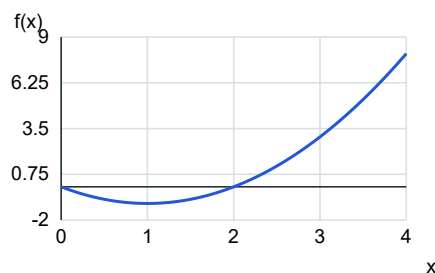
☐ C 9

☐ D 0

Jacques, 9th ed. · §4.2 · Exercise 4.2 2(a) · printed p. 286 / PDF p. 303

Q8 · Rank 3 / Application

Evaluate $f'(x)$ for $f(x) = x^2 - 2x$ at $x = 3$.



☐ A 3

☐ B 4

☐ C 6

☐ D -4

Jacques, 9th ed. · §4.2 · Exercise 4.2 2(b) · printed p. 286 / PDF p. 303

Lesson 06 · Power, constant, sum and difference rules

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

Evaluate $f'(x)$ for $f(x) = x^3 - 4x^2 + 2x - 8$ at $x = 0$.

☐ A -8

☐ B 0

☐ C 2

☐ D -6

Jacques, 9th ed. · §4.2 · Exercise 4.2 2(c) · printed p. 286 / PDF p. 303

Q10 · Rank 4 / Challenge · OPTIONAL

Evaluate $f'(x)$ for $f(x) = \sqrt{x} - \frac{2}{x}$ at $x = 4$.

☐ A $\frac{1}{8}$

☐ B $\frac{1}{2}$

☐ C $\frac{3}{2}$

☐ D $\frac{3}{8}$

Jacques, 9th ed. · §4.2 · Exercise 4.2 2(e) · printed p. 286 / PDF p. 303

Lesson 07 · Chain and product rules

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Use the chain rule to differentiate $y = (5x + 1)^3$.

☐ A $15(5x + 1)^2$

☐ B $3(5x + 1)^2$

☐ C $15(5x + 1)^3$

☐ D $5(5x + 1)^2$

Jacques, 9th ed. · §4.4 · Exercise 4.4 1(a) · printed p. 312 / PDF p. 329

Q2 · Rank 1 / Foundation

Use the chain rule to differentiate $y = (x + 9)^5$.

☐ A $5x^4$

☐ B $5(x + 9)^4$

☐ C $9(x + 9)^4$

☐ D $5(x + 9)^5$

Jacques, 9th ed. · §4.4 · Exercise 4.4 1(c) · printed p. 312 / PDF p. 329

Lesson 07 · Chain and product rules

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

Use the chain rule to differentiate $y = (2x - 7)^8$.

☐ A $8(2x - 7)^7$

☐ B $16(2x - 7)^8$

☐ C $16(2x - 7)^7$

☐ D $-56(2x - 7)^7$

Jacques, 9th ed. · §4.4 · Exercise 4.4 1(b) · printed p. 312 / PDF p. 329

Q4 · Rank 2 / Core

Use the chain rule to differentiate $y = (4x^2 - 7)^3$.

☐ A $3(4x^2 - 7)^2$

☐ B $12x(4x^2 - 7)^2$

☐ C $24x(4x^2 - 7)^3$

☐ D $24x(4x^2 - 7)^2$

Jacques, 9th ed. · §4.4 · Exercise 4.4 1(d) · printed p. 312 / PDF p. 329

Lesson 07 · Chain and product rules

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

Use the product rule to differentiate $y = x(3x + 4)^2$.

☐ A $(3x + 4)(9x + 4)$

☐ B $6x(3x + 4)$

☐ C $(3x + 4)^2$

☐ D $6(3x + 4)$

Jacques, 9th ed. · §4.4 · Exercise 4.4 2(a) · printed p. 312 / PDF p. 329

Q6 · Rank 2 / Core

Use the product rule to differentiate $y = x^2(x - 2)^3$.

☐ A $6x(x - 2)^2$

☐ B $x(x - 2)^2(5x - 4)$

☐ C $2x(x - 2)^3$

☐ D $3x^2(x - 2)^2$

Jacques, 9th ed. · §4.4 · Exercise 4.4 2(b) · printed p. 312 / PDF p. 329

Lesson 07 · Chain and product rules

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

Differentiate $y = (5x + 7)^2$ by using the chain rule.

☐ **A** $10x + 14$

☐ **B** $50x + 14$

☐ **C** $50x + 70$

☐ **D** $25x + 70$

Jacques, 9th ed. · §4.4 · Exercise 4.4 4(a) · printed p. 312 / PDF p. 329

Q8 · Rank 3 / Application

Differentiate $y = x^5(x + 2)^2$ by using the product rule.

☐ **A** $10x^4(x + 2)$

☐ **B** $5x^4(x + 2)^2$

☐ **C** $2x^5(x + 2)$

☐ **D** $x^4(x + 2)(7x + 10)$

Jacques, 9th ed. · §4.4 · Exercise 4.4 5(a) · printed p. 312 / PDF p. 329

Lesson 07 · Chain and product rules

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

Find an expression for marginal revenue when the demand function is $P = (100 - Q)^3$.

☐ A $MR = (100 - Q)^2(100 - 4Q)$

☐ B $MR = -3(100 - Q)^2$

☐ C $MR = (100 - Q)^3$

☐ D $MR = (100 - Q)^2(100 + 2Q)$

Jacques, 9th ed. · §4.4 · Exercise 4.4 6(a) · printed p. 312 / PDF p. 329

Q10 · Rank 4 / Challenge · OPTIONAL

Use the quotient rule to differentiate $y = \frac{x}{x-5}$, where $x \neq 5$.

☐ A $\frac{5}{(x-5)^2}$

☐ B $-\frac{5}{(x-5)^2}$

☐ C 1

☐ D $\frac{1}{x-5}$

Jacques, 9th ed. · §4.4 · Exercise 4.4 3(a) · printed p. 312 / PDF p. 329

Lesson 08 · Derivatives of exponentials and logarithms

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Write down the derivative of $y = e^{6x}$.

☐ A e^{6x}

☐ B $6e^x$

☐ C $6e^{6x}$

☐ D $e^{6x}/6$

Jacques, 9th ed. · §4.8 · Exercise 4.8 1(a) · printed p. 370 / PDF p. 387

Q2 · Rank 1 / Foundation

Write down the derivative of $y = e^{-342x}$.

☐ A $342e^{-342x}$

☐ B e^{-342x}

☐ C $-342e^x$

☐ D $-342e^{-342x}$

Jacques, 9th ed. · §4.8 · Exercise 4.8 1(b) · printed p. 370 / PDF p. 387

Lesson 08 · Derivatives of exponentials and logarithms

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

Write down the derivative of $y = \ln(3x)$, where $x > 0$.

☐ A $1/x$

☐ B $3/x$

☐ C $1/(3x)$

☐ D $\ln 3$

Jacques, 9th ed. · §4.8 · Exercise 4.8 3(a) · printed p. 370 / PDF p. 387

Q4 · Rank 2 / Core

Write down the derivative of $y = 2e^{-x} + 4e^x$.

☐ A $2e^{-x} + 4e^x$

☐ B $-2e^{-x} + 4e^x$

☐ C $-2e^{-x} - 4e^x$

☐ D $-e^{-x} + e^x$

Jacques, 9th ed. · §4.8 · Exercise 4.8 1(c) · printed p. 370 / PDF p. 387

Lesson 08 · Derivatives of exponentials and logarithms

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

Write down the derivative of $y = 10e^{4x} - 2x^2 + 7$.

☐ A $10e^{4x} - 4x$

☐ B $40e^{4x} - 4x + 7$

☐ C $40e^{4x} - 4x$

☐ D $40e^x - 4x$

Jacques, 9th ed. · §4.8 · Exercise 4.8 1(d) · printed p. 370 / PDF p. 387

Q6 · Rank 2 / Core

Write down the derivative of $y = \ln(-13x)$, where $x < 0$.

☐ A $-13/x$

☐ B $-1/x$

☐ C $1/(-13x)$

☐ D $1/x$

Jacques, 9th ed. · §4.8 · Exercise 4.8 3(b) · printed p. 370 / PDF p. 387

Lesson 08 · Derivatives of exponentials and logarithms

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

If USD 4000 is saved in an account offering a return of 4% compounded continuously, the future value, S , after t years is given by $S = 4000e^{0.04t}$. Write down an expression for dS/dt and hence find the exact value of the rate of growth after five years.

☐ A $160e^{0.2}$

☐ B $4000e^{0.2}$

☐ C $0.04e^{0.2}$

☐ D $160e^{0.04}$

Jacques, 9th ed. · §4.8 · Exercise 4.8 2(2) · printed p. 370 / PDF p. 387

Q8 · Rank 3 / Application

If USD 4000 is saved in an account offering a return of 4% compounded continuously, the future value, S , after t years is given by $S = 4000e^{0.04t}$. Calculate S at $t = 5$ and $t = 5.01$ and hence estimate the rate of growth at $t = 5$. Use unrounded intermediate balances; round the rate to two decimal places.

☐ A 1.95

☐ B 195.46

☐ C 195.42

☐ D 4885.61

Jacques, 9th ed. · §4.8 · Exercise 4.8 2(1)(a)–(b) and hence · printed p. 370 / PDF p. 387

Lesson 08 · Derivatives of exponentials and logarithms

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

Use the product rule to differentiate $y = x \ln x$. Assume $x > 0$.

☐ A 1

☐ B $x + \ln x$

☐ C $\ln x + 1$

☐ D $\ln x + 1/x$

Jacques, 9th ed. · §4.8 · Exercise 4.8 5(b) · printed p. 370 / PDF p. 387

Q10 · Rank 4 / Challenge · OPTIONAL

Use the chain rule to differentiate $y = \ln(x^4 + 3x^2)$. Assume $x \neq 0$.

☐ A $1/(x^4 + 3x^2)$

☐ B $4x^3 + 6x$

☐ C $1/x$

☐ D $\frac{4x^3+6x}{x^4+3x^2}$

Jacques, 9th ed. · §4.8 · Exercise 4.8 4(b) · printed p. 370 / PDF p. 387

Lesson 09 · Marginal quantities and local estimates

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

If the demand function is $P = 100 - 4Q$, find an expression for marginal revenue in terms of Q .

☐ A $100 - 4Q$

☐ B $100 - 8Q$

☐ C $100Q - 4Q^2$

☐ D -4

Jacques, 9th ed. · §4.3 · Exercise 4.3 1, MR · printed p. 301 / PDF p. 318

Q2 · Rank 1 / Foundation

If the average cost function is $AC = 15/Q + 2Q + 9$, what are the fixed costs?

☐ A 9

☐ B 0

☐ C 15

☐ D 2

Jacques, 9th ed. · §4.3 · Exercise 4.3 4, fixed costs · printed p. 302 / PDF p. 319

Lesson 09 · Marginal quantities and local estimates

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

A firm has production function $Q = 50L - 0.01L^2$, where L denotes the workforce. Find the marginal product of labour when $L = 10$.

☐ A 499☐ B 48☐ C 50☐ D 49.8

Jacques, 9th ed. · §4.3 · Exercise 4.3 5(b) · printed p. 302 / PDF p. 319

Q4 · Rank 2 / Core

If the demand function is $P = 100 - 4Q$, find an expression for total revenue in terms of Q .

☐ A $100Q - 4Q^2$ ☐ B $100 - 4Q$ ☐ C $100Q - 4Q$ ☐ D $100 - 8Q$

Jacques, 9th ed. · §4.3 · Exercise 4.3 1, TR · printed p. 301 / PDF p. 318

Lesson 09 · Marginal quantities and local estimates

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

If $AC = 15/Q + 2Q + 9$, write down an expression for the marginal cost function.

☐ A $2 - 15/Q^2$

☐ B $4Q + 9$

☐ C $2Q + 9$

☐ D $15 + 2Q^2 + 9Q$

Jacques, 9th ed. · §4.3 · Exercise 4.3 4, MC · printed p. 302 / PDF p. 319

Q6 · Rank 2 / Core

The fixed costs of a good are 100 and the variable costs are $2 + Q/10$ per unit. Find an expression for marginal cost.

☐ A $2 + Q/10$

☐ B $100 + 2Q + Q^2/10$

☐ C $2 + Q/5$

☐ D $100 + Q/5$

Jacques, 9th ed. · §4.3 · Exercise 4.3* 3(a), MC · printed p. 303 / PDF p. 320

Lesson 09 · Marginal quantities and local estimates

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

If $P = 100 - 4Q$, estimate the change in total revenue brought about by a 0.3-unit increase in output from a current level of 12 units.

☐ A 0.84☐ B 4☐ C 1.2☐ D 15.6

Jacques, 9th ed. · §4.3 · Exercise 4.3 1, final estimate · printed p. 301 / PDF p. 318

Q8 · Rank 3 / Application

Fixed costs are 100 and variable costs are $2 + Q/10$ per unit. Evaluate MC at $Q = 30$ and hence estimate the change in TC for a 2-unit increase in output from 30. Choose the estimated change.

☐ A 8☐ B 16.4☐ C 10☐ D 16

Jacques, 9th ed. · §4.3 · Exercise 4.3* 3(b) · printed p. 303 / PDF p. 320

Lesson 09 · Marginal quantities and local estimates

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

Total cost is $TC = 250 + 20Q$. Marginal revenue is 18 at $Q = 219$. If production increases slightly from 219, would you expect profit to increase, decrease or stay the same?

- ☐ A Decrease: marginal profit is -2 .
- ☐ B Increase: marginal profit is 2.
- ☐ C Stay the same: fixed cost is constant.
- ☐ D Increase: marginal profit is 18.

Jacques, 9th ed. · §4.3 · Exercise 4.3* 7 · printed p. 303 / PDF p. 320

Q10 · Rank 4 / Challenge · OPTIONAL

A firm has demand $P = 100 - 4\sqrt{Q} - 3Q$. Estimate the change in TR when output rises by 0.25 units from 9, and compare it with the exact change. Choose (estimate, exact change), with the exact change rounded to two decimal places.

- ☐ A (7, 7)
- ☐ B (28, 6.78)
- ☐ C (7, 6.78)
- ☐ D (6.78, 7)

Jacques, 9th ed. · §4.3 · Exercise 4.3* 1(c) · printed p. 302 / PDF p. 319

Lesson 10 · Stationary points and extrema

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Find and classify the stationary point of $y = -x^2 + x + 1$.

☐ **A** Maximum at $(1/2, 5/4)$.

☐ **B** Minimum at $(1/2, 5/4)$.

☐ **C** Maximum at $(1, 1)$.

☐ **D** Minimum at $(-1/2, 1/4)$.

Jacques, 9th ed. · §4.6 · Exercise 4.6 1(a) · printed p. 345 / PDF p. 362

Q2 · Rank 1 / Foundation

Find and classify the stationary point of $y = x^2 - 4x + 4$.

☐ **A** Maximum at $(2, 0)$.

☐ **B** Minimum at $(4, 4)$.

☐ **C** Minimum at $(2, 0)$.

☐ **D** Minimum at $(0, 4)$.

Jacques, 9th ed. · §4.6 · Exercise 4.6 1(b) · printed p. 345 / PDF p. 362

Lesson 10 · Stationary points and extrema

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

Find and classify the stationary point of $y = x^2 - 20x + 105$.

☐ **A** Minimum at (20, 105).

☐ **B** Maximum at (10, 5).

☐ **C** Minimum at (10, 105).

☐ **D** Minimum at (10, 5).

Jacques, 9th ed. · §4.6 · Exercise 4.6 1(c) · printed p. 345 / PDF p. 362

Q4 · Rank 2 / Core

Find and classify the stationary points of $y = -x^3 + 3x$.

☐ **A** Minimum $(-1, -2)$; maximum $(1, 2)$.

☐ **B** Maximum $(-1, -2)$; minimum $(1, 2)$.

☐ **C** Minimum $(0, 0)$; maximum $(1, 2)$.

☐ **D** Minimum $(-1, 2)$; maximum $(1, -2)$.

Jacques, 9th ed. · §4.6 · Exercise 4.6 1(d) · printed p. 345 / PDF p. 362

Lesson 10 · Stationary points and extrema

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

The demand function is $P = 40 - 2Q$. Find the value of Q which maximises total revenue.

☐ A 20☐ B 10☐ C 40☐ D 0

Jacques, 9th ed. · §4.6 · Exercise 4.6 2 · printed p. 345 / PDF p. 362

Q6 · Rank 2 / Core

The average cost function is $AC = 13/Q + Q + 2$. Find the value of Q which minimises average cost, for $Q > 0$.

☐ A 13☐ B $2\sqrt{13}$ ☐ C $\sqrt{13}$ ☐ D $-\sqrt{13}$

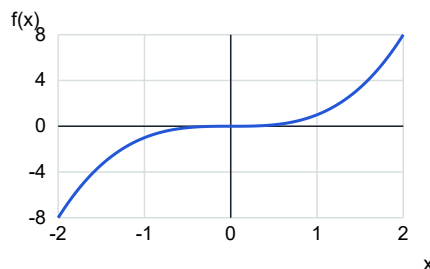
Jacques, 9th ed. · §4.6 · Exercise 4.6 4, differentiation · printed p. 345 / PDF p. 362

Lesson 10 · Stationary points and extrema

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

For $f(x) = x^3$, verify that $f'(0) = f''(0) = 0$ and classify the stationary point at 0.



☐ A Local maximum.

☐ B Local minimum.

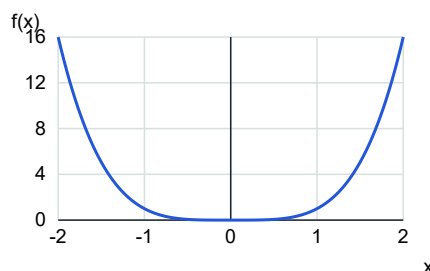
☐ C Both a maximum and a minimum.

☐ D Stationary point of inflection.

Jacques, 9th ed. · §4.6 · Exercise 4.6* 2(a) · printed p. 347 / PDF p. 364

Q8 · Rank 3 / Application

For $f(x) = x^4$, verify that $f'(0) = f''(0) = 0$ and classify the stationary point at 0.



☐ A Local minimum.

☐ B Local maximum.

☐ C Stationary point of inflection.

☐ D No stationary point.

Jacques, 9th ed. · §4.6 · Exercise 4.6* 2(b) · printed p. 347 / PDF p. 364

Lesson 10 · Stationary points and extrema

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

For $f(x) = -x^6$, verify that $f'(0) = f''(0) = 0$ and classify the stationary point at 0.

☐ A Local minimum.

☐ B Local maximum.

☐ C Stationary point of inflection.

☐ D No stationary point.

Jacques, 9th ed. · §4.6 · Exercise 4.6* 2(c) · printed p. 347 / PDF p. 364

Q10 · Rank 4 / Challenge · OPTIONAL

Daily sales are $S = t^3 - 24t^2 + 180t + 60$ for $0 \leq t \leq 13$. Find the maximum and minimum daily sales during the period between $t = 5$ and $t = 9$. Choose (maximum, minimum).

☐ A (492, 460)

☐ B (485, 465)

☐ C (492, 465)

☐ D (465, 492)

Jacques, 9th ed. · §4.6 · Exercise 4.6* 4(c) · printed p. 347 / PDF p. 364

Lesson 11 · Profit optimisation and feasible decisions

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

A firm has demand $P = aQ + b$ ($a < 0$, $b > 0$), fixed costs c and variable costs d per unit. Write down a general expression for total revenue.

☐ A $aQ + b$

☐ B $aQ^2 + bQ$

☐ C $aQ^2 + b$

☐ D $c + dQ$

Jacques, 9th ed. · §4.7 · Exercise 4.7* 1(a), TR · printed p. 359 / PDF p. 376

Q2 · Rank 1 / Foundation

A firm has demand $P = aQ + b$ ($a < 0$, $b > 0$), fixed costs c and variable costs d per unit. Write down a general expression for total cost.

☐ A $c + d$

☐ B $cQ + d$

☐ C $c + dQ$

☐ D dQ

Jacques, 9th ed. · §4.7 · Exercise 4.7* 1(a), TC · printed p. 359 / PDF p. 376

Lesson 11 · Profit optimisation and feasible decisions

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

A firm has demand $P = aQ + b$ ($a < 0$, $b > 0$), fixed costs c and variable costs d per unit. Deduce the marginal cost.

☐ A c ☐ B $c + d$ ☐ C dQ ☐ D d

Jacques, 9th ed. · §4.7 · Exercise 4.7* 1(b), MC · printed p. 359 / PDF p. 376

Q4 · Rank 2 / Core

A firm has demand $P = aQ + b$ ($a < 0$, $b > 0$), fixed costs c and variable costs d per unit. Deduce the marginal revenue.

☐ A $2aQ + b$ ☐ B $aQ + b$ ☐ C $2a + b$ ☐ D $aQ^2 + bQ$

Jacques, 9th ed. · §4.7 · Exercise 4.7* 1(b), MR · printed p. 359 / PDF p. 376

Lesson 11 · Profit optimisation and feasible decisions

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

A firm has demand $P = aQ + b$ ($a < 0$, $b > 0$), fixed costs c and variable costs d per unit. Assume also $b > d \geq 0$ and $c \geq 0$. Find the non-negative output which maximises profit.

☐ A $\frac{b-d}{2a}$

☐ B $\frac{d-b}{2a}$

☐ C $\frac{d-b}{a}$

☐ D $\frac{-b}{2a}$

Jacques, 9th ed. · §4.7 · Exercise 4.7* 1(c) · printed p. 359 / PDF p. 376

Q6 · Rank 2 / Core

A firm has $TC = Q^2 + 50Q + 10$ and demand $P = 200 - 4Q$. Find the level of output needed to maximise profit.

☐ A 25

☐ B 30

☐ C 15

☐ D 50

Jacques, 9th ed. · §4.6 · Exercise 4.6* 6(a) · printed p. 347 / PDF p. 364

Lesson 11 · Profit optimisation and feasible decisions

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

A firm has demand $P = 60 - 0.5Q$, fixed costs 10 and variable costs $Q + 3$ per unit. Find the maximum profit.

☐ A 541.5☐ B 531.5☐ C 1083☐ D 19

Jacques, 9th ed. · §4.6 · Exercise 4.6* 1 · printed p. 347 / PDF p. 364

Q8 · Rank 3 / Application

The average cost per person of hiring a tour guide for a week's river cruise is $AC = 3Q^2 - 192Q + 3500$, where $0 < Q \leq 30$. Find the minimum average cost for the trip.

☐ A 428☐ B 3500☐ C 32☐ D 440

Jacques, 9th ed. · §4.6 · Exercise 4.6 9 · printed p. 346 / PDF p. 363

Lesson 11 · Profit optimisation and feasible decisions

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

A firm has production $Q = 3\sqrt{L}$, sells everything at USD 50 per unit, and pays USD 10 per unit of labour. Ignore other costs. Find the value of L which maximises profit, treating labour as continuous.

☐ A 56.25☐ B 7.5☐ C 225☐ D 0

Jacques, 9th ed. · §4.6 · Exercise 4.6.8 · printed p. 346 / PDF p. 363

Q10 · Rank 4 / Challenge · OPTIONAL

A firm has $TC = Q^2 + 50Q + 10$ and $P = 200 - 4Q$. A tax of t dollars per good is added to its costs. For $0 \leq t < 150$, the firm continues to maximise profit. By how much does the price increase?

☐ A t ☐ B $t/5$ ☐ C $2t/5$ ☐ D $3t/5$

Jacques, 9th ed. · §4.6 · Exercise 4.6* 6(b) · printed p. 347 / PDF p. 364

Lesson 12 · Signed demand elasticity and revenue

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

If an airline increases prices for business class flights by 8%, demand falls by about 2.5%. Estimate the signed elasticity of demand.

☐ A 0.3125☐ B -3.2☐ C -0.3125☐ D -2.5

Jacques, 9th ed. · §4.5 · Exercise 4.5 4(a), estimate · printed p. 326 / PDF p. 343

Q2 · Rank 1 / Foundation

Find the signed price elasticity of demand at $P = 6$ for $P = 30 - 2Q$.

☐ A -0.25☐ B -4☐ C 0.25☐ D -0.5

Jacques, 9th ed. · §4.5 · Exercise 4.5 3(a) · printed p. 326 / PDF p. 343

Lesson 12 · Signed demand elasticity and revenue

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

Find the signed price elasticity of demand at $P = 6$ for $P = 30 - 12Q$.

☐ A $-1/12$

☐ B -1.5

☐ C 0.25

☐ D -0.25

Jacques, 9th ed. · §4.5 · Exercise 4.5 3(b) · printed p. 326 / PDF p. 343

Q4 · Rank 2 / Core

If the demand equation is $Q + 4P = 60$, find a general expression for the signed price elasticity in terms of P , for $0 < P < 15$.

☐ A -4

☐ B $-P/(15 - P)$

☐ C $P/(15 - P)$

☐ D $-(15 - P)/P$

Jacques, 9th ed. · §4.5 · Exercise 4.5* 3, general expression · printed p. 327 / PDF p. 344

Lesson 12 · Signed demand elasticity and revenue

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

If the demand equation is $Q + 4P = 60$, for what value of P is demand unit elastic?

☐ A 15☐ B 30☐ C 7.5☐ D 0

Jacques, 9th ed. · §4.5 · Exercise 4.5* 3, unit elasticity · printed p. 327 / PDF p. 344

Q6 · Rank 2 / Core

Find the signed elasticity of demand in terms of Q for $P = 20 - 0.05Q$, for $0 < Q < 400$.

☐ A $-(400 - Q)/Q$ ☐ B -0.05 ☐ C $-Q/(400 - Q)$ ☐ D $+(400 - Q)/Q$

Jacques, 9th ed. · §4.5 · Exercise 4.5 6(a) · printed p. 327 / PDF p. 344

Lesson 12 · Signed demand elasticity and revenue

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

For demand $P = 20 - 0.05Q$, at what value of Q is demand unit elastic?

☐ A 400☐ B 20☐ C 10☐ D 200

Jacques, 9th ed. · §4.5 · Exercise 4.5 6(b) · printed p. 327 / PDF p. 344

Q8 · Rank 3 / Application

For demand $P = 20 - 0.05Q$, find marginal revenue and evaluate it where demand is unit elastic. Choose (MR , value at that point).

☐ A $(20 - 0.05Q, 10)$ ☐ B $(20 - 0.1Q, 0)$ ☐ C $(20Q - 0.05Q^2, 2000)$ ☐ D $(20 - 0.1Q, 20)$

Jacques, 9th ed. · §4.5 · Exercise 4.5 6(c) · printed p. 327 / PDF p. 344

Lesson 12 · Signed demand elasticity and revenue

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

An airline raises business class prices by 8% and demand falls by about 2.5%. Is demand elastic, inelastic or unit elastic?

- ☐ Elastic, because price rose more than demand fell.
- ☐ Unit elastic, because both quantities changed.
- ☒ Inelastic, because $|E| = 0.3125 < 1$.
- ☐ Undefined, because elasticity is negative.

Jacques, 9th ed. · §4.5 · Exercise 4.5 4(a), classification · printed p. 326 / PDF p. 343

Q10 · Rank 4 / Challenge · OPTIONAL

Demand is $Q = 1000/P^2$. Estimate the percentage change in demand when price increases by 2% from 5, then compare it with the exact change when price rises from 5 to 5.1. Choose (estimate, exact), rounded to two decimal places.

- ☒ A (−4%, −3.88%)
- ☐ B (−2%, −3.88%)
- ☐ C (−4%, −4%)
- ☐ D (4%, 3.88%)

Jacques, 9th ed. · §4.5 · Exercise 4.5 5(a)–(b) · printed p. 326 / PDF p. 343

Lesson 13 · Antiderivatives and recovering totals

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Find $\int 6x^5 dx$.

☐ **A** $6x^6 + C$

☐ **B** $x^6 + C$

☐ **C** $30x^4 + C$

☐ **D** $x^5 + C$

Jacques, 9th ed. · §6.1 · Exercise 6.1 1(a) · printed p. 496 / PDF p. 513

Q2 · Rank 1 / Foundation

Find $\int x^4 dx$.

☐ **A** $4x^3 + C$

☐ **B** $x^5 + C$

☐ **C** $x^5/5 + C$

☐ **D** $x^4/4 + C$

Jacques, 9th ed. · §6.1 · Exercise 6.1 1(b) · printed p. 496 / PDF p. 513

Lesson 13 · Antiderivatives and recovering totals

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

Find $\int (2x^3 - 6x) dx$.

☐ A $2x^4 - 6x^2 + C$

☐ B $6x^2 - 6 + C$

☐ C $x^4/2 - 6x^2 + C$

☐ D $x^4/2 - 3x^2 + C$

Jacques, 9th ed. · §6.1 · Exercise 6.1 1(f) · printed p. 496 / PDF p. 513

Q4 · Rank 2 / Core

Find $\int (x^2 - 8x + 3) dx$.

☐ A $x^3/3 - 4x^2 + 3x + C$

☐ B $x^3 - 8x^2 + 3x + C$

☐ C $x^3/3 - 4x^2 + 3 + C$

☐ D $2x - 8 + C$

Jacques, 9th ed. · §6.1 · Exercise 6.1 1(g) · printed p. 496 / PDF p. 513

Lesson 13 · Antiderivatives and recovering totals

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

Find $\int (ax + b) dx$, where a and b are constants.

☐ **A** $ax^2 + bx + C$

☐ **B** $ax^2/2 + bx + C$

☐ **C** $ax^2/2 + b + C$

☐ **D** $a + C$

Jacques, 9th ed. · §6.1 · Exercise 6.1 1(h) · printed p. 496 / PDF p. 513

Q6 · Rank 2 / Core

Find total cost if marginal cost is $MC = Q + 5$ and fixed costs are 20.

☐ **A** $Q + 25$

☐ **B** $Q^2/2 + 5Q$

☐ **C** $Q^2/2 + 5Q + 20$

☐ **D** $Q^2 + 5Q + 20$

Jacques, 9th ed. · §6.1 · Exercise 6.1 2(a) · printed p. 496 / PDF p. 513

Lesson 13 · Antiderivatives and recovering totals

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

Marginal cost is $MC = 2Q + 6$. Total cost is 212 when $Q = 8$. Find total cost when $Q = 14$.

☐ A 280☐ B 416☐ C 28☐ D 380

Jacques, 9th ed. · §6.1 · Exercise 6.1 3 · printed p. 496 / PDF p. 513

Q8 · Rank 3 / Application

Find the total revenue and demand functions corresponding to $MR = 20 - 2Q$. Use $TR(0) = 0$ and $Q > 0$ for the demand function. Choose (TR, P) .

☐ A $(20Q - Q^2, 20 - Q)$ ☐ B $(20Q - Q^2, 20 - 2Q)$ ☐ C $(20 - 2Q, 20/Q - 2)$ ☐ D $(20Q - 2Q^2, 20 - 2Q)$

Jacques, 9th ed. · §6.1 · Exercise 6.1 4(a) · printed p. 496 / PDF p. 513

Lesson 13 · Antiderivatives and recovering totals

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

Find the consumption function if the marginal propensity to consume is 0.6 and consumption is 10 when income is 5. Deduce the savings function. Choose $(C(Y), S(Y))$.

☐ **A** $(0.6Y + 10, 0.4Y - 10)$

☐ **B** $(0.6Y + 7, 0.4Y - 7)$

☐ **C** $(0.6Y + 7, 0.4Y + 7)$

☐ **D** $(0.6Y, 0.4Y)$

Jacques, 9th ed. · §6.1 · Exercise 6.1 5 · printed p. 496 / PDF p. 513

Q10 · Rank 4 / Challenge · OPTIONAL

Differentiate $F(x) = (2x + 1)^5$. Hence find $\int (2x + 1)^4 dx$.

☐ **A** $(2x + 1)^5/5 + C$

☐ **B** $2(2x + 1)^5/5 + C$

☐ **C** $(2x + 1)^5/10 + C$

☐ **D** $8(2x + 1)^3 + C$

Jacques, 9th ed. · §6.1 · Exercise 6.1 8(1) · printed p. 496 / PDF p. 513

Lesson 14 · Definite integrals, area and accumulation

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Evaluate the definite integral $\int_0^2 x^3 dx$.

☐ A 4

☐ B 8

☐ C 16

☐ D 0

Jacques, 9th ed. · §6.2 · Exercise 6.2 3(a) · printed p. 509 / PDF p. 526

Q2 · Rank 1 / Foundation

Evaluate the definite integral $\int_{-2}^2 x^3 dx$.

☐ A 8

☐ B 0

☐ C 4

☐ D -8

Jacques, 9th ed. · §6.2 · Exercise 6.2 3(b) · printed p. 509 / PDF p. 526

Lesson 14 · Definite integrals, area and accumulation

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

Find the exact area under the curve $y = (x - 2)^2$ between $x = 2$ and $x = 3$.

☐ A 1

☐ B 3

☐ C $\frac{1}{3}$

☐ D 0

Jacques, 9th ed. · §6.2 · Exercise 6.2 2(b) · printed p. 509 / PDF p. 526

Q4 · Rank 2 / Core

Evaluate the integral $\int_1^3 4x^2 dx$.

☐ A 36

☐ B 108

☐ C $\frac{26}{3}$

☐ D $\frac{104}{3}$

Jacques, 9th ed. · §6.2 · Exercise 6.2 1(a) · printed p. 509 / PDF p. 526

Lesson 14 · Definite integrals, area and accumulation

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

Evaluate the integral $\int_1^2 (4x^3 - 3x^2 + 4x + 2) dx$.

☐ A 16

☐ B 20

☐ C 12

☐ D -16

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Q6 · Rank 2 / Core

Evaluate the integral $\int_{-1}^2 (5x^2 - 4x + 6) dx$.

☐ A $\frac{23}{3}$

☐ B 27

☐ C $\frac{52}{3}$

☐ D -27

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Lesson 14 · Definite integrals, area and accumulation

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

Find the exact area under the curve $y = 2x^2 + x + 3$ between $x = 1$ and $x = 5$.

☐ A $\frac{320}{3}$

☐ B $\frac{334}{3}$

☐ C $\frac{248}{3}$

☐ D 53

Jacques, 9th ed. · §6.2 · Exercise 6.2 2(a) · printed p. 509 / PDF p. 526

Q8 · Rank 3 / Application

Write down an expression for $g'(x)$ when $g(x) = \int_2^x (5t^2 - 2t) dt$.

☐ A $10x - 2$

☐ B $5x^2 - 2x - 16$

☐ C $5x^2 - 2x$

☐ D $\frac{5}{3}x^3 - x^2$

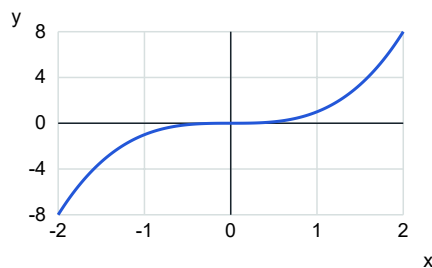
Jacques, 9th ed. · §6.2 · Exercise 6.2* 10 · printed p. 511 / PDF p. 528

Lesson 14 · Definite integrals, area and accumulation

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

What is the actual area between the x axis and the graph of $y = x^3$, over the range $-2 \leq x \leq 2$?

☐ A 0☐ B 4☐ C -8☐ D 8

Jacques, 9th ed. · §6.2 · Exercise 6.2 3, final follow-up · printed p. 509 / PDF p. 526

Q10 · Rank 4 / Challenge · OPTIONAL

Evaluate the integral $\int_2^3 \frac{2}{x^3} dx$.

☐ A $-\frac{5}{36}$ ☐ B $\frac{1}{9}$ ☐ C $\frac{5}{36}$ ☐ D $\frac{19}{2}$

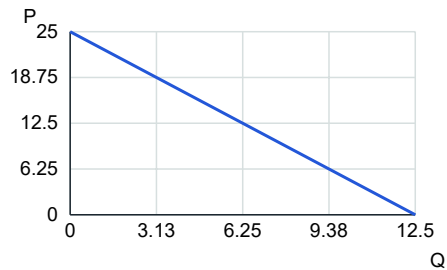
Jacques, 9th ed. · §6.2 · Exercise 6.2 1(b) · printed p. 509 / PDF p. 526

Lesson 15 · Consumer and producer surplus

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

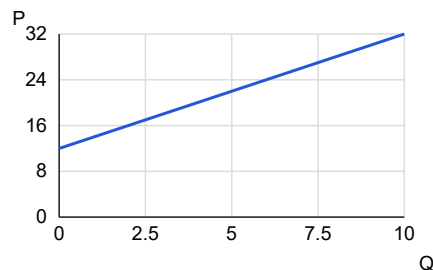
Find the consumer's surplus at $P = 5$ for the demand function $P = 25 - 2Q$.

☐ A 50☐ B 100☐ C 200☐ D 10

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Q2 · Rank 1 / Foundation

Find the producer's surplus at $Q = 9$ for the supply function $P = 12 + 2Q$.

☐ A 270☐ B 189☐ C 81☐ D 162

Jacques, 9th ed. · §6.2.2 · Exercise 6.2 5(a) · printed p. 509 / PDF p. 526

Lesson 15 · Consumer and producer surplus

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

Find the consumer's surplus at $Q = 8$ for the demand function $P = 50 - 4Q$.

☐ A 144☐ B 256☐ C 16☐ D 128

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Q4 · Rank 2 / Core

Find the consumer's surplus for $P = 50 - 2Q - 0.01Q^2$ when $Q = 10$. Give the exact value.

☐ A $320/3$ ☐ B 100☐ C 290☐ D $1190/3$

Jacques, 9th ed. · §6.2.1 · Exercise 6.2 6(a) · printed p. 509 / PDF p. 526

Lesson 15 · Consumer and producer surplus

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

The producer's surplus at $Q = a$ for the supply function $P = 6 + 8Q$ is 400. Find a , with $a \geq 0$.

☐ A 5☐ B 10☐ C 20☐ D -10

Jacques, 9th ed. · §6.2.2 · Exercise 6.2* 2(b) · printed p. 510 / PDF p. 527

Q6 · Rank 2 / Core

What is the consumer's surplus at $P = 75$ when demand is $P = 100 - Q^2$?

☐ A $75/3$ ☐ B $125/3$ ☐ C $225/3$ ☐ D $250/3$

Jacques, 9th ed. · §6.2.1 · Chapter 6 Multiple Choice Questions Question 10 · printed p. 517 / PDF p. 534

Lesson 15 · Consumer and producer surplus

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

Demand is $P = -Q_D^2 - 4Q_D + 68$ and supply is $P = Q_S^2 + 2Q_S + 12$. Find the consumer's surplus assuming pure competition.

☐ A 144☐ B $224/3$ ☐ C $176/3$ ☐ D $400/3$

Jacques, 9th ed. · §6.2.1 · Exercise 6.2 7(a) · printed p. 510 / PDF p. 527

Q8 · Rank 3 / Application

Demand is $P = -Q_D^2 - 4Q_D + 68$ and supply is $P = Q_S^2 + 2Q_S + 12$. Find the producer's surplus assuming pure competition.

☐ A $224/3$ ☐ B 144☐ C $176/3$ ☐ D $400/3$

Jacques, 9th ed. · §6.2.2 · Exercise 6.2 7(b) · printed p. 510 / PDF p. 527

Lesson 15 · Consumer and producer surplus

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

Demand is $P = 74 - Q_D^2$ and supply is $P = (Q_S + 2)^2$. Calculate the consumer's and producer's surplus under pure competition. Choose (CS, PS) .

☐ A (250/3, 400/3)

☐ B (400/3, 250/3)

☐ C (245, 245)

☐ D (250/3, 650/3)

Jacques, 9th ed. · §6.2.1–6.2.2 · Exercise 6.2* 3 · printed p. 511 / PDF p. 528

Q10 · Rank 4 / Challenge · OPTIONAL

Demand is $P = 80 - 6\sqrt{Q}$. Find the change in consumer's surplus due to a decrease in price from 62 to 56.

☐ A 128

☐ B 54

☐ C -74

☐ D 74

Jacques, 9th ed. · §6.2.1 · Exercise 6.2* 14(a) · printed p. 512 / PDF p. 529

Lesson 16 · Investment flows and accumulated decisions

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

If the investment flow of a company is $I(t) = 8000t^{1/3}$, calculate the capital formation during the first eight years.

☐ A 96 000☐ B 128 000☐ C 76 800☐ D 256 000

Jacques, 9th ed. · §6.2.3 · Chapter 6 Multiple Choice Questions Question 5 · printed p. 516 / PDF p. 533

Q2 · Rank 1 / Foundation

Given investment flow $I(t) = 2400\sqrt{t}$, calculate total capital formation during the first four years.

☐ A 19 200☐ B 12 800☐ C 6400☐ D 4800

Jacques, 9th ed. · §6.2.3 · Exercise 6.2 9(a) · printed p. 510 / PDF p. 527

Lesson 16 · Investment flows and accumulated decisions

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

For a green spaces project, investment is $I(t) = 0.6t^2 + 10$ million dollars per year for the first five years, then $J(t) = 25e^{0.4(5-t)}$ for $5 \leq t \leq 10$. Verify $I(5) = J(5)$. What is their common value?

☐ A 10 million dollars per year.

☐ B 75 million dollars.

☐ C 25 million dollars per year.

☐ D 25 million dollars.

Jacques, 9th ed. · §6.2.3 · Exercise 6.2* 13(a), boundary verification · printed p. 512 / PDF p. 529

Q4 · Rank 2 / Core

If investment flow is $I(t) = 5000t^{1/4}$, calculate capital formation from the end of the second year to halfway through the fifth year. Give the answer to the nearest whole number.

☐ A 20 000

☐ B 20 393

☐ C 16 703

☐ D 9500

Jacques, 9th ed. · §6.2.3 · Exercise 6.2 8 · printed p. 510 / PDF p. 527

Lesson 16 · Investment flows and accumulated decisions

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

Find an expression for capital formation between $t = 0$ and $t = T$ if net investment is $I(t) = At^a$, where A and a are positive constants.

☐ A AT^a

☐ B $AT^{a+1}/(a+1)$

☐ C AT^{a+1}

☐ D AaT^{a-1}

Jacques, 9th ed. · §6.2.3 · Exercise 6.2* 7(a) · printed p. 511 / PDF p. 528

Q6 · Rank 2 / Core

Find an expression for capital formation between $t = 0$ and $t = T$ if net investment is $I(t) = Ae^{at}$, where A and a are positive constants.

☐ A Ae^{aT}

☐ B $A(e^{aT} - 1)$

☐ C Ae^{aT}/a

☐ D $A(e^{aT} - 1)/a$

Jacques, 9th ed. · §6.2.3 · Exercise 6.2* 7(b) · printed p. 511 / PDF p. 528

Lesson 16 · Investment flows and accumulated decisions

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

An oil company starts extraction at $t = 0$. The extraction rate, in thousands of barrels per year, is $dN/dt = 30t^2 - 4t^3$. Calculate the number of barrels extracted during the first four years.

☐ A 384 barrels.

☐ B 384,000 barrels.

☐ C 96,000 barrels.

☐ D 192,000 barrels.

Jacques, 9th ed. · §6.2.3 · Exercise 6.2 11 · printed p. 510 / PDF p. 527

Q8 · Rank 3 / Application

Given investment flow $I(t) = 2400\sqrt{t}$, find an expression for annual capital formation during the N th year and hence the first year in which annual capital formation exceeds USD 4000. Choose the first year.

☐ A Year 1.

☐ B Year 3.

☐ C Year 4.

☐ D Year 5.

Jacques, 9th ed. · §6.2.3 · Exercise 6.2 9(b) · printed p. 510 / PDF p. 527

Lesson 16 · Investment flows and accumulated decisions

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

If net investment is $I(t) = 100e^{0.1t}$, calculate capital formation from the end of the second year to the end of the fifth year. Round to two decimal places.

☐ A 648.72☐ B 427.32☐ C 221.40☐ D 164.87

Jacques, 9th ed. · §6.2.3 · Exercise 6.2* 6(a) · printed p. 511 / PDF p. 528

Q10 · Rank 4 / Challenge · OPTIONAL

Calculate the present value of a continuous revenue stream of USD 1000 per year at a 5% discount rate, paid for 3 years. Use continuous discounting and round to two decimal places.

☐ A 3000.00☐ B 2591.51☐ C 2785.84☐ D 3153.42

Jacques, 9th ed. · §6.2.3 · Exercise 6.2* 8(a) · printed p. 511 / PDF p. 528